
Geometry Matters: Comparing GCN and SchNet for HOMO–LUMO Gap Prediction on QM9

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Abstract

Predicting molecular electronic properties from structure is a central problem in computational chemistry, where data-driven surrogates promise orders-of-magnitude speedups over density functional theory (DFT). We compare two graph-based neural networks on the HOMO–LUMO gap regression task from the QM9 dataset: a Graph Convolutional Network (GCN) baseline that operates on bond-connectivity graphs, and SchNet, a continuous-filter convolutional network that incorporates 3D interatomic distances. We hypothesize that 3D geometric information is essential for this task because orbital energies depend on spatial atomic arrangement, not just bond topology. Trained on 110,000 molecules, SchNet achieves a test MAE of 51.7 meV ($R^2 = 0.996$), a $3.4\times$ improvement over GCN’s 177.8 meV ($R^2 = 0.961$). A learning-curve analysis across training-set sizes from 5k to 110k shows that the SchNet–GCN gap *widens* with more data, indicating that SchNet’s geometric inductive bias does not merely speed up early learning but raises the achievable error ceiling.

1 Introduction

Quantum-mechanical methods such as density functional theory (DFT) provide accurate predictions of molecular electronic properties but are computationally expensive, often requiring hours of CPU time per molecule. As a result, screening even modestly sized chemical libraries quickly becomes infeasible. Data-driven surrogate models that map molecular structure directly to target properties offer a route to accelerate materials discovery, drug design, and the rational design of organic photovoltaics by orders of magnitude [Gilmer et al., 2017, Schütt et al., 2017].

The QM9 dataset [Ramakrishnan et al., 2014], drawn from the GDB-17 chemical universe [Ruddigkeit et al., 2012], contains approximately 130,000 small organic molecules (up to 9 heavy atoms among C, N, O, F) with 19 DFT-computed regression targets. Among these, the HOMO–LUMO gap $\Delta\epsilon$ —the energy difference between the highest occupied and lowest unoccupied molecular orbitals—is of particular practical importance: it governs optical absorption, photochemical reactivity, and charge-transport behavior in organic electronic devices. Accurate, fast surrogates for $\Delta\epsilon$ would enable virtual screening of candidate chromophores and semiconductor materials at a scale that is currently impossible with first-principles simulation.

A central modeling question is what structural information the network should be exposed to. Molecules can be represented purely as bond-connectivity graphs, but the HOMO–LUMO gap depends sensitively on the 3D arrangement of atoms because orbital overlap is a function of interatomic distance. In this report we compare a baseline graph convolutional network [Kipf and Welling, 2017], which sees only bond topology, against SchNet [Schütt et al., 2017, 2018], which incorporates continuous interatomic distances via radial-basis-function filters. Our central hypothesis is that the geometric inductive bias should produce a substantial accuracy gain. As a secondary question, we

ask how this gap scales with training data: do both models close the gap with enough examples, or does the geometric bias raise the achievable ceiling?

2 Methods

Baseline model: Graph Convolutional Network (GCN). We use the GCN of Kipf and Welling [Kipf and Welling, 2017] as the baseline. GCN is one of the architectures explicitly covered in this course and is a standard graph-data baseline. Each layer aggregates features from a node’s bonded neighbors using a symmetrically-normalized adjacency matrix and applies a learned linear transformation followed by a nonlinearity:

$$H^{(l+1)} = \sigma\left(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H^{(l)} W^{(l)}\right), \quad (1)$$

where $\tilde{A} = A + I$ adds self-loops and $\tilde{D}$ is its degree matrix. We stack 3 GCN layers with hidden dimension 128, a global-mean readout over atoms, and a 2-layer MLP head producing the scalar $\Delta\varepsilon$ prediction (51,201 trainable parameters). Atom features are the 11-dimensional categorical features supplied by PyTorch Geometric. GCN is a sensible baseline because it operates natively on molecular graphs, has been widely applied to chemistry [Gilmer et al., 2017], and—critically—ignores 3D coordinates entirely, making it a clean control for testing whether geometric information matters.

Variant: SchNet. SchNet [Schütt et al., 2017, 2018] is a continuous-filter convolutional network proposed for modeling quantum interactions. Like GCN, it follows the message-passing paradigm; unlike GCN, its messages between atoms i and j are modulated by a continuous filter $W(\|\mathbf{r}_i - \mathbf{r}_j\|)$ produced from a radial-basis-function (RBF) expansion of the interatomic distance:

$$\mathbf{x}_i^{(l+1)} = \mathbf{x}_i^{(l)} + \sum_{j \in \mathcal{N}(i)} \mathbf{x}_j^{(l)} \circ W^{(l)}(\|\mathbf{r}_i - \mathbf{r}_j\|), \quad (2)$$

where the neighborhood $\mathcal{N}(i)$ is defined by a 10 Å distance cutoff rather than by chemical bonds. We use the standard PyTorch Geometric implementation with 6 interaction blocks, hidden dimension 128, 50 RBFs, and a 10 Å cutoff (455,809 trainable parameters).

Why this variant. SchNet differs from GCN along one well-defined axis: it conditions messages on continuous 3D distances rather than discrete bond connectivity. This is precisely the inductive bias one would expect to matter for HOMO–LUMO gap prediction, where orbital overlap and therefore frontier-orbital energies are determined by spatial geometry. Comparing GCN to SchNet thus isolates the contribution of 3D geometric information on a quantum-property regression task. SchNet was published at NeurIPS 2017 [Schütt et al., 2017] and is not on the list of architectures covered in lecture, satisfying the variant requirement.

Training setup. We use a random 110k/10k/10k train/validation/test split with a fixed seed, following the convention of the SchNet paper. Targets are normalized to zero mean and unit variance over the training set; reported metrics are computed on the original (eV) scale. Both models are trained with Adam [Kingma and Ba, 2015] (initial lr 10^{-3} , ReduceLROnPlateau on validation MAE with patience 8 and factor 0.5) using batch size 128 and L1 loss. GCN was trained for 200 epochs and SchNet for 150 epochs, with early stopping (patience 25 and 20, respectively) on validation MAE. Both runs converged before triggering early stopping. Implementation uses PyTorch [Paszke et al., 2019] and PyTorch Geometric [Fey and Lenssen, 2019]; experiments were conducted on a single Tesla T4 GPU.

3 Results and Discussion

Evaluation metric. Following standard QM9 practice [Gilmer et al., 2017, Schütt et al., 2017], we report Mean Absolute Error (MAE) on the test set in the original physical units of electron-volts:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|. \quad (3)$$

MAE is appropriate because the target distribution is approximately symmetric and unimodal, the chemistry literature reports MAEs on this benchmark, and the metric is interpretable as the average error in the physical quantity of interest. We additionally report R^2 as a unitless complement.

Test-set performance. Table 1 reports test MAE and R^2 for both models trained on the full 110k-molecule training split. SchNet achieves a $3.4\times$ reduction in test MAE compared to GCN (51.7 meV vs. 177.8 meV), a 71% relative improvement. The corresponding R^2 values (0.996 vs. 0.961) confirm that SchNet captures essentially all of the variance in the held-out gap distribution, while GCN leaves a meaningful residual. SchNet’s 51.7 meV result is competitive with the value originally reported by Schütt et al. [Schütt et al., 2017] (~ 63 meV) on the same target.

Table 1: Test-set performance on QM9 HOMO–LUMO gap prediction (110k training molecules).

Model	Test MAE (meV) ↓	R^2 ↑	Parameters	Train time
GCN (baseline)	177.8	0.961	51,201	58 min
SchNet (variant)	51.7	0.996	455,809	92 min

Training curves. Figure 1 plots validation MAE versus training epoch for both models. SchNet reaches GCN’s converged validation error within roughly 10 epochs and continues to improve, ultimately settling near 50 meV while GCN plateaus near 175 meV. The two curves remain well-separated throughout training, indicating that the gap is not a transient artifact of optimization speed but reflects a genuine difference in achievable accuracy.

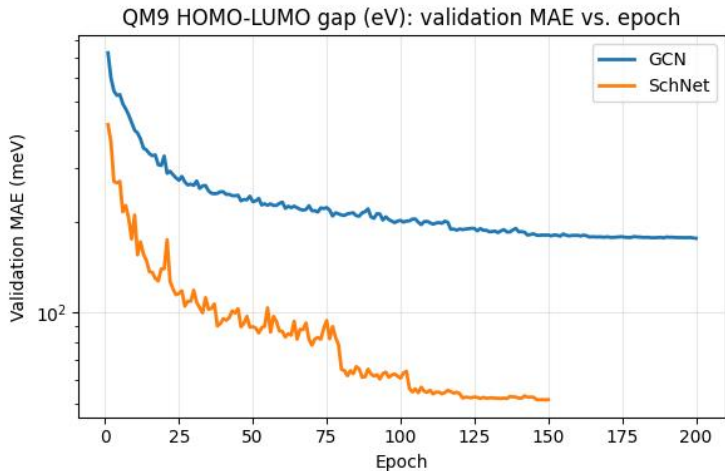


Figure 1: Validation MAE (meV, log scale) versus training epoch for GCN and SchNet on QM9 HOMO–LUMO gap prediction. SchNet reaches GCN’s converged error within ~ 10 epochs and continues to improve, settling roughly an order of magnitude below GCN’s plateau.

Hypothesis check. The hypothesis from Section 2 is strongly supported: replacing topological message passing with distance-conditioned message passing reduces test MAE by 71%. Because GCN and SchNet share the same overall message-passing skeleton and were trained with identical optimizer, schedule, loss, and data splits, the gap is most plausibly attributed to the 3D-geometric inductive bias rather than to optimizer or capacity differences alone—though SchNet does have roughly $9\times$ more parameters, an issue we revisit through the learning-curve analysis below.

Additional analysis: learning curves. To investigate whether the SchNet advantage is primarily due to its geometric inductive bias or simply to its larger capacity, we retrained both models on training subsets of size 5k, 20k, and 50k molecules (in addition to the 110k full run), holding the validation and test splits fixed. Figure 2 and Table 2 show the result.

The result is opposite to the standard “inductive-bias helps most in the low-data regime” prediction. Both models follow approximately power-law learning curves, but SchNet’s slope is steeper: SchNet’s relative advantage grows from $1.83\times$ at $N=5k$ to $3.44\times$ at $N=110k$. Two observations follow. First, even at $N=5k$, SchNet already outperforms GCN trained on $22\times$ as much data (273.6 meV vs. GCN’s 283.3 meV at 20k), so the geometric bias does provide substantial sample efficiency in absolute terms. Second, however, GCN appears to be *capacity-limited*, not merely data-limited: even with the full 110k training set it plateaus near ~ 178 meV, while SchNet keeps improving. This

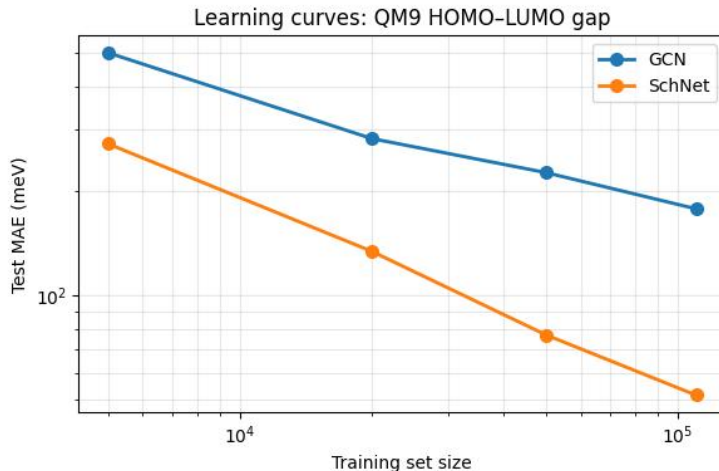


Figure 2: Test MAE (meV) as a function of training set size (log-log) for GCN and SchNet. Both models improve roughly as a power law with N , but SchNet’s slope is steeper, so the gap between the two architectures *widens* with more data.

Table 2: Test MAE and SchNet/GCN ratio across training-set sizes.

Train size N	GCN MAE (meV)	SchNet MAE (meV)	GCN/SchNet ratio
5,000	500.6	273.6	$1.83\times$
20,000	283.3	134.1	$2.11\times$
50,000	226.0	77.0	$2.93\times$
110,000	177.8	51.7	$3.44\times$

suggests that bond-topology features alone cannot represent the function from molecular structure to HOMO–LUMO gap with arbitrary accuracy, no matter how much training data is provided. The 3D-geometric inductive bias raises the achievable error ceiling, not merely the convergence rate. This argument also responds to the parameter-count concern raised earlier: extra capacity alone cannot explain why GCN’s curve flattens—it flattens because the input representation is missing information that SchNet’s input contains.

4 Conclusion

We compared a topology-only GCN baseline against geometry-aware SchNet on QM9 HOMO–LUMO gap prediction. SchNet substantially outperformed GCN on test MAE (51.7 meV vs. 177.8 meV, a $3.4\times$ reduction), directly supporting the hypothesis that 3D interatomic distances carry information that bond connectivity alone does not capture for this electronic property. A learning-curve analysis across training sizes from 5k to 110k showed, somewhat counterintuitively, that the SchNet–GCN gap *widens* with more data: GCN’s accuracy plateaus near ~ 178 meV regardless of training size, while SchNet’s continues to improve. We interpret this as evidence that the geometric inductive bias raises the achievable error ceiling, not just the convergence rate—bond topology is fundamentally insufficient for this task.

The most surprising finding was the failure of the standard inductive-bias-as-sample-efficiency story: we expected SchNet’s advantage to be most pronounced when data was scarce, but instead found GCN to be capacity-limited even at full scale. Given more time and compute, the natural next step would be to evaluate models that incorporate *angular* information in addition to distances, such as DimeNet [Klicpera et al., 2020], to test whether the marginal gain from angles is comparable to the gain from distances reported here. A second direction would be to repeat this comparison across the other 18 QM9 targets to characterize which properties benefit most strongly from geometric inductive bias—we would predict the largest gains on properties whose physics is most directly tied to spatial

geometry (e.g. dipole moment, atomization energy) and the smallest gains on properties that are well-determined by composition alone (e.g. heat capacity).

Collaboration Statement

This is a solo project. I designed the experiment (variant selection, learning-curve analysis, hyperparameter choices, data splits), implemented and ran all training, and wrote up the analysis and interpretation of results, including the revised reading of the learning-curve data once I saw the gap was widening rather than closing with more data. I used Claude as a debugging and sanity-check assistant throughout. Specifically, it helped me work through environment issues during setup a CUDA architecture mismatch for a Tesla P100 that the preinstalled PyTorch did not support, and a missing `torch_cluster` dependency that SchNet’s radius-graph builder needed and helped me check my LaTeX template and reference setup once my results were in. I also used it for occasional rubber-ducking on whether my interpretation of the learning-curve trend was sound. All experimental decisions, the running of the training jobs, and the final writing are my own.

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